

CADILLAC AREA GROUNDWATER CONTAMINATION PUBLIC MEETING

August 7, 1989

Wexford County Courthouse

Cadillac, Michigan



Panel - DIANNE ROYCRAFT, MDNR
GARY SIMONS, MDNR
STEVE SANDERS, USEPA
REBECCA DILWORTH, E.C. Jordan
ELIZABETH DeSWARTE, E.C. Jordan
TED TAYLOR, E.C. Jordan

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1 MS. DIANNE ROYCRAFT: My name is Dianne Roycraft. I
2 work with the Michigan Department of Natural Resources, the
3 Environmental Response Division here in Cadillac. And part
4 of my responsibilities include project manager for several
5 groundwater investigation and clean-up projects, one of
6 which is the Cadillac Area Groundwater Contamination
7 Project, which brings us here tonight. I'd like to start
8 by first introducing our panel and after which I'll provide
9 a brief overview of the project. And the investigation and
10 feasibility study which was conducted will be presented in
11 more detail by those who did conduct that. And the meeting
12 is to provide the form for the public to comment on the
13 work that has been done. And comments will be received at
14 this time tonight as well as throughout the public comment
15 period which does extend through August 28th.

16 The panel includes tonight, on my far right,
17 Gary Simons, who works in the Environmental Response
18 Division of the Department of Natural Resources in the
19 Lansing Office. Gary was a project manager for the
20 Northernaire Plating Facility project. Next to Gary is
21 Steve Sanders, who is the remedial project manager from the
22 U.S. Environmental Protection Agency, Region 5 in Chicago.
23 The next three, Rebecca Dilworth, Elizabeth DeSwarte and
24 Ted Taylor, are from the company E.C. Jordan, and they were
25 the prime contractor in implementing the remedial

1 investigation and feasibility study.

2 Since 1978 a number of residential water supply wells
3 in the Cadillac area were found to be contaminated with
4 various industrial organic compounds. And these compounds
5 were detected. Wells were subsequently abandoned. The
6 Michigan Environmental Response Act, Act 307, in 1982, did
7 create a fund for replacement of residential wells as well
8 as investigations and clean-up activities. And, through
9 that Michigan Environmental Response Fund, the potable
10 water was provided to those businesses and homes that were
11 impacted. And, also, this fund was used for conducting the
12 remedial investigation and feasibility study with the
13 objective of fully defining the problem as well as
14 identifying the available remedies.

15 Because two Superfund sites exist within the study
16 area, this investigation, though State-funded, was aligned
17 closely with the Federal Superfund Program, according to
18 the EPA protocol set forth in the Superfund Program.
19 Again, tonight's meeting is to provide a forum for public
20 comment and specifically on the clean-up options that have
21 been identified. And, in order to adjust the time of our
22 presentations, I would like to have an indication of the
23 number of comments that we might expect to receive tonight,
24 if I can have a show of hands at this time. Okay. Good.

25 I would like to mention, also, that we have provided a

1 sign-up sheet here in front. And, if you could please sign
2 up so that we do have, for the public record, your correct
3 name and corporation, we'd appreciate that. And the
4 comments need to be made from the front rather than from
5 your seat. The public comment does extend through
6 August 28th, and I did provide an agenda which is available
7 as you walk in on the right. And at the bottom does
8 indicate the person who needs to receive those public
9 comments, being Dan O'Riordan, mail code 5PA-14. Dan is a
10 community relations coordinator with USEPA Office of Public
11 Affairs. And his phone number is also provided.

12 The remedial investigation and feasibility study does
13 incorporate basically the extent of the industrial park
14 area with the general boundaries being -- everyone see this
15 okay? The study area boundaries include -- this being our
16 north direction -- Leeson Avenue on the west, the Ann Arbor
17 railroad tracks on the east, Wright Street on the south and
18 the -- basically Grahek Street on the north, so we are
19 north of 13th Street. The two Superfund sites within that
20 study area, again, being Northernnaire and proposed
21 Superfund facility Kysor of Cadillac. And within the study
22 area is the Cadillac Municipal well field.

23 Again, the purpose of the investigation was to define
24 the extent, lateral and vertical extent, of groundwater and
25 soils contamination and identify the feasible remedies that

1 are available, cost-effective, environmentally sound and
2 socially acceptable remedial alternatives. Within the
3 remedial investigation, the nature, extent and probable
4 sources of contamination were the objectives.

5 And, with that brief introduction, I would like to
6 introduce Ted Taylor, who will be presenting more details
7 on the investigation itself. And Ted was on site and very
8 intensively involved in that investigation. Ted.

9
10 MR. TED TAYLOR: Good evening. What I'd like to do is
11 just pick up where Dianne left off. Like she said, I work
12 with E.C. Jordan Company at the Farmington Hills, Michigan,
13 office with a home office in Portland, Maine. And we've
14 been fairly active as subcontractors to the DNR since the
15 early 1980's starting with the Northernaire facility. Like
16 Dianne mentioned, in 1978 they were having complaints about
17 groundwater contamination in private supply wells. And
18 some emergency response actions were taken with some
19 drilling and testing involved for contaminants in
20 groundwater.

21 And, since that time, we've -- in separate drilling
22 phases, we've been out to investigate the site and have a
23 fairly good picture of the groundwater contamination that
24 exists in approximately a one-mile or greater square
25 mile -- one-square-mile area of the industrial park. And,

1 of course, the primary concern has been all along is: How
2 are these contaminants going to affect the city well field?
3 The first phase, like I said, concentrated on the
4 Northernnaire facility where Hexavalent Chromium, due to
5 their operations out there, were detected. So I'll go
6 through that plume first.

7 What these base maps here show are the identified
8 plumes that we've found out in the industrial park and
9 I'll discuss briefly. Just by putting in monitoring wells
10 with drilling rigs and testing the water every --
11 periodically as we drilled, we would test for contaminants.
12 And we were able to define the limits of both volatile
13 organic compound plumes, which are generally used as
14 solvents by industries, and also Hexavalent Chromium, which
15 was used in the metal plating facility by Northernnaire.
16 Northernnaire facility is right here. Leeson Avenue is
17 right here. And 6th Street runs this way, just for
18 reference (indicating).

19 The Hexavalent Chromium plume we've determined was
20 mainly out in the undeveloped area of the industrial park,
21 the city well field being here (indicating). Groundwater
22 flow in the intermediate aquifer, which we refer to in this
23 area, was to the north and then starts to go to the
24 northwest as it comes out in this area of industrial park
25 (indicating). So it posed no real threat to the well

1 fields at that time. And so we continued to investigate
2 volatile organic plumes, which we knew existed out there.
3 And, since plumes are intermingling -- there are several
4 sources out there and contamination was widespread -- we
5 realized that the most effective way to clean the plumes up
6 would be simultaneously. We could wait on this plume and
7 clean it up with the rest.

8 So essentially what we did was, through three or four
9 different drilling phases, put in monitoring wells at all
10 corners of the industrial park. We concentrated wells in
11 areas, you know, both ourselves, and plus some of the
12 potentially responsible parties did some of their own
13 investigations. And there's probably close to 120, 130 or
14 even more monitoring wells out there for the purpose of
15 defining the contamination. Then, again, we would put some
16 wells in by the City well field, too, to see what the
17 picture looked like there.

18 Essentially what you have are three aquifers in the
19 area, aquifer being a water-bearing unit that you can --
20 that supports -- that can give you a large supply of water.
21 And the Cadillac well field is down around 300 feet or so
22 deep. They're pumping from that kind of depth.
23 Fortunately the contamination isn't at that depth yet.
24 Most of this contamination here is either in a water table
25 aquifer, which is from 20 to 50 feet below ground surface,

1 and intermediate aquifer, which is from 50 to, say, 200,
2 250 feet and separated by a clay unit. The City well field
3 is pumping below that.

4 So knowing -- or having a head start with knowing
5 Northernnaire, we realized other potentially responsible
6 parties included, I think, are pretty well identified here.
7 Northernnaire, Kysor has been identified. And then, like I
8 said, it's so widespread you start getting a lot of
9 other -- the plumes start intermingling. We believe that
10 the Four Winns have contributed some type of contamination
11 in this area, which is another plume that's within the
12 larger plume, emanating in this direction. Groundwater
13 flow, of course, is in this direction. That's why the
14 plume spreads out along that path (indicating).

15 Another potentially responsible party was the old Four
16 Star facility. I believe they're no longer there. It's
17 another -- Micad, or some other facility is there now. And
18 another one that has come up recently is the Rexair
19 facility. And they're currently doing their own work to
20 continue investigation at that area. So shallow
21 contamination is what is shown on this map. You have --
22 again, emanating from soils behind the Kysor facility,
23 there are plumes of trichlorethylene and other solvent-type
24 materials in the groundwater out to this point
25 (indicating).

1 And this line here marks where the clay aquitard,
2 which is a layer that restricts groundwater flow
3 vertically, stops at the industrial park. And this is all
4 one large aquifer out here, 200, 250 feet or so. And here
5 is a shallower aquifer where a lot of the contamination
6 rests above a clay layer (indicating). So you can see it's
7 fairly widespread in this area, the shallow contamination.
8 And then this represents the contamination the next aquifer
9 down, about 150 feet or so.

10 Some of the contamination in the shallow aquifer has
11 leaked through that clay layer, and that is what we believe
12 is causing the contamination at this depth here. This is
13 volatile organic contamination. Again, this is the
14 Hex-Chrome plume, which rests right on top of the clay
15 layer. There were two weak plumes we detected out in the
16 North Park subdivision. They're at 50 to -- I think this
17 area here is -- the depth of the plume is about 50 feet
18 deep, and out here it's about 140 feet deep (indicating).
19 Again, they're relatively small and relatively well removed
20 from the municipal well field.

21 The final potentially responsible party that we've
22 been able to identify is the Rexair facility, where you can
23 see a very concentrated plume has been detected, again, of
24 trichlorethylene, a solvent. And of primary concern is it
25 is upgrade, hydrogeologically, of the Cadillac well field.

1 It's still vertically above where the city wells are
2 pumping from, as far as we know at this point. But it is
3 headed in that direction. The contaminants can be sinking,
4 one, due to their nature; two, due to the pumping in the
5 Cadillac well field. And so I believe emergency response
6 has been started by the DNR to insure that the
7 contamination isn't going to get to those wells for now.
8 And, until something more permanent can be put in place as
9 far as remediation, the emergency response will handle
10 this.

11 So, again, just to sum up, we've, I guess, identified
12 up to seven plumes, which I've shown on both the front and
13 back, with as many as five potentially responsible parties.
14 And this is based on the result of the field work that
15 we've done. Jordan got involved, I believe, in '83 or '84,
16 in that time frame. But, like I say, other studies had --
17 Kysor had done some studies before that. The DNR did some
18 studies out by Four Star. And then since then we've been
19 keeping pace with working with DNR to try to put the pieces
20 together.

21 MS. ROYCRAFT: I would clarify that the emergency
22 response action that Ted referred to as being implemented
23 by the DNR for the Rexair plume actually was implemented by
24 Rexair.

25 MR. TAYLOR: Oh, okay.

1 MS. ROYCRAFT: Becky Dilworth is now going to present
2 the findings and the methodology for the feasibility study
3 that was conducted by Jordan.
4

5 MS. REBECCA DILWORTH: Good evening. At the
6 conclusion of the remedial investigation that Ted was
7 talking about, a risk assessment is conducted to identify
8 the migration pathways, the receptors and the contaminants
9 of concern, those which pose a significant risk. Once all
10 that's identified, we then start the feasibility study.
11 And the purpose of the feasibility study, as Dianne has
12 mentioned, is to identify remedial action objectives that
13 can address all of these problems and the remedial action
14 objectives and the alternatives that we'll address as
15 objectives.

16 The remedial action objectives are statements of the
17 proposed level of cleanup associated with specific
18 chemical concentrations detected at the site. We try to
19 identify levels that are protective of human health and the
20 environment. And those are based on the contaminants of
21 concern and those exposure routes and the receptors; who is
22 drinking the water, who is having a direct contact threat
23 to contaminated soils.

24 At the Kysor and Northernnaire site, with regards to
25 groundwater, the exposure pathway identified as presenting

1 a potential threat to public health is the domestic use of
2 the groundwater; drinking, bathing, cooking, showering.
3 Therefore, the general response objectives for each of
4 those plumes was to reduce that public health risk. We
5 then developed groundwater target cleanup levels, which
6 are actual concentration levels that indicate a protective
7 level. And those are based on regulatory standards or
8 calculated levels.

9 Because the contaminated soils at the Kysor site are
10 subsurface, at this point there's no direct contact threat.
11 It's really seen as a problem because there's continual
12 leaching or potential leaching of the contaminants to the
13 groundwater. So that's how we assess that site and develop
14 target cleanup levels based on those contaminants'
15 leachability and the concentrations at which they were
16 detected.

17 The first step in the whole process is to -- in
18 developing those alternatives that will obtain those
19 response objectives, those numerical standards that are
20 identified in the FS is to identify and screen technologies
21 that are specific to that Cadillac Industrial Park, to the
22 site conditions and to the contaminants that exist there.
23 We have to insure that the technologies are suitable for
24 that location and for those conditions. Once we develop or
25 we identify a whole list of technologies and we screen

1 those, and it obviously becomes a smaller list, and then
2 we combine those technologies into alternatives.

3 Because there are Superfund sites within the study
4 area, we're obviously, as Dianne said, doing this under the
5 Superfund protocol. The Superfund Amendments and
6 Re-authorization Act emphasizes the use of technologies
7 that reduce the toxicity mobility and volume of the
8 contaminants as opposed to just preventing exposure. I
9 mean, we could go in there and say, "Okay. We won't let
10 you drink any of the water, and then you'll be protected
11 from having exposure." But the real goal is to get rid of
12 the contamination, reduce the toxicity mobility and volume.

13 For the groundwater and developing the alternatives
14 for the groundwater, the plumes that Ted identified, those
15 different-colored plumes, were identified as one study area
16 to be remediated by one alternative. Even though there are
17 different responsible parties, we didn't think it was
18 practical to go in there and remediate each plume
19 separately. We look -- because they're so intermingled, we
20 identified it as one alternative.

21 In the beginning phases of developing all the
22 alternatives, we looked at a no-action alternative and
23 four treatment alternatives. The no-action alternative is
24 developed as a baseline alternative to compare the other
25 alternatives to. And it consists solely of groundwater

1 monitoring for changes in the contaminant concentrations
2 over time. Beth will talk a little bit about more details
3 of those alternatives.

4 The groundwater treatments that we looked at involve
5 extraction of the groundwater, some form of Hexavalent
6 Chromium treatment, -- the Hexavalent Chromium plume is
7 from the Northernnaire site -- some form of volatile organic
8 contaminant compound treatment -- they have to be treated
9 separately -- and then a discharge of that groundwater.
10 Once you treat it, you've got to put it somewhere. So we
11 looked at different discharge options and came up with
12 discharging to the Clam River once it's treated to the
13 required treatment levels.

14 We take the whole array of alternatives at that point,
15 and we do an initial screening of those. We screen against
16 their effectiveness, their implementability and cost
17 factors. But that's an initial screen. Once you've gone
18 through that screen, then you'll come up with a smaller
19 amount of alternatives that then go through a very detailed
20 analysis.

21 I just want to show you how the groundwater
22 alternatives work at this site. You can see that we
23 have -- here's the no-action alternative here, an
24 extraction, which is a component of every single -- every
25 treatment alternative. We then have two choices of a

1 Hexavalent Chromium treatment; ion exchange or carbon
2 adsorption. You would choose one of those.

3 Then we have three potential treatments for the
4 volatile organic compounds. And those are listed here; UV
5 oxidation, air-stripping with carbon adsorption and steam
6 stripping. And then they all involve a discharge to the
7 Clam River. This overhead is taken directly from the
8 feasibility study and just summarizes what those
9 alternatives are. It's kind of hard to see it in the flow
10 chart. But, again, we have the no-action, then each
11 alternative with different components in each one. And
12 then this just shows what the potential options for the
13 Hexavalent Chromium treatment. Again, Beth is going to
14 talk about the details of those, how they work, what the
15 costs are associated with those.

16 The soil remedial alternatives were developed in a
17 similar way. And they include a range from a no-action
18 alternative, again, that's used as a baseline, to simply
19 containment so that there is no leachability, to actual
20 treatment alternatives. These -- we came up again. We
21 identify technologies. We screen those technologies based
22 on the site specific characteristics and the waste
23 specific characteristics, took those technologies to form
24 alternatives, screen those again and came up with a list to
25 undergo the detailed analysis: No-action, surficial

1 capping with a subsurface containment wall and then the
2 three treatment -- four treatment systems that have been --
3 will undergo the detailed analysis.

4 When we take these alternatives, we're analyzing them
5 against nine criteria. This slide indicates what those
6 nine criteria are:

7 The overall protection. How is this proposed
8 alternative going to reduce the current and future threats
9 to human health and the environment? Can this alternative
10 do this?

11 Compliance with applicable or relevant and
12 appropriate requirements. Those are the regulatory
13 standards, both Federal and State. Will they meet these
14 requirements? Will they meet the required treatment
15 levels? When we discharge to the Clam River, will this
16 alternative meet those required standards? Will we meet
17 all of the other environmental types of regulations that
18 we're required to?

19 Long-term effectiveness is the measure of the
20 long-term effectiveness of an alternative in protecting
21 human health and the environment long after the design
22 objectives have been met. Is there any residual risk, are
23 some of the questions. If an alternative might clean up
24 the problem, but there will be residual risk, we may not
25 consider that one. What are the long-term management

1 considerations? And what is the reliability of controls
2 over time?

3 The fourth criteria is reduction of toxicity,
4 mobility or volume. As I mentioned, that is required. And
5 it addresses the technical performances of the treatment
6 technologies under the proposed alternative. And, again,
7 it talks about whether -- the types and quantity of
8 hazardous contaminant residuals resulting from the
9 treatment process.

10 The short-term effectiveness talks a little bit about:
11 What's the effect on the community while we're
12 implementing this? Will the workers be at risk while this
13 is being implemented? And how will the community be
14 impacted?

15 Implementability addresses the technical and
16 administrative feasibility of the alternatives including
17 the availability of the equipment, the demonstrative
18 performance of the equipment and the time it takes to
19 implement.

20 Cost is obviously a criteria and addresses the capital
21 costs, which include direct and indirect costs and
22 operation and maintenance costs of an alternative. The
23 costs that we're going to be presenting to you tonight are
24 actually the present worth costs. And those evaluate --
25 that number that you're going to be seeing is the cost that

1 occurs over a time frame by discounting the future cost to
2 the present. So it's how much it's going to cost us now
3 taking into account all the costs we're going to accrue
4 over the time it takes to clean up the site.

5 And also acceptance by the State and acceptance by the
6 community is considered. And that's why your public
7 comments tonight until the end of the comment period are
8 very important to the final decision.

9 Now I'm going to turn it over to Beth, who's going to
10 talk about the actual details of the alternatives.

11
12 MS. ELIZABETH DeSWARTE: Good evening. It's nice to
13 have a full house here tonight. And what I would like to
14 do at this time is to give you a little more insight in all
15 these mystery technologies and alternatives we've been
16 hearing about.

17 First of all, I'd like to go over the groundwater
18 alternatives that were conducted through detailed analysis
19 for the site. And we passed through detailed analysis,
20 four alternatives. And, as Becky mentioned, the first
21 alternative we consider is the no-action. And that is the
22 alternative with which we use as a baseline to look at the
23 other alternatives. And, for the no-action alternative
24 for groundwater, it consists primarily of groundwater
25 monitoring, and that is to monitor for those contaminants

1 of concern and see if and how they migrate.

2 And also along with no-action alternative is a
3 five-year site review. And after five years is a review
4 looking at the data and seeing if there are possible
5 additional things that could be done at the site.

6 For our second alternative -- well, I should say for
7 all of the alternatives, as Becky mentioned, we go through
8 an extraction. And then for the Hexavalent Chromium we
9 can perform either ion exchange or carbon adsorption. And
10 then we go into a volatile organic treatment and then to a
11 discharge.

12 So what I'll do -- what will be easiest is to go over
13 right up front the Hexavalent Chromium treatments, and then
14 we can go over the overall alternatives. And what we
15 consider for Hexavalent Chromium are ion exchange and
16 carbon adsorption. And the ion exchange process works on
17 the principle of: You have a resin bed. It has a charged
18 resin that will attract, in this case, the Hexavalent
19 Chromium ion, which has an opposing charge. And that
20 Hexavalent Chromium ion will be attracted to the resin and
21 adhere to it. And then once all the resin sites are taken
22 up, we can regenerate that and then start the process all
23 over again.

24 So we considered that process. One hindering factor
25 to that is it is often hard to separate out -- the resin

1 doesn't always recognize which are Hex-Chrome ions and
2 which are other ions of the same charge. Therefore, this
3 alternative, it has somewhat of a higher cost due to a
4 great deal of regeneration waste that's produced every
5 year. And our guesstimates are that there would be
6 approximately 200,000 gallons per year of a hazardous
7 material that would have to be disposed.

8 And for the second Hexavalent Chromium alternative
9 that we looked at, that included carbon adsorption. And
10 that's similar to an ion exchange. You have an activated
11 carbon material in a packed column. And the carbon has
12 available sites. It's similar to a sponge that has little
13 intricate holes in the -- the volatile -- or excuse me --
14 the Hexavalent Chromium ions will attach or adsorb onto the
15 activated carbon. And, with this system, we don't get
16 quite as many -- or quite the volume of waste as we do
17 with the ion exchange process.

18 Okay. Now, looking at alternative two, for volatile
19 organic treatment we have the ultraviolet and oxidation
20 treatment. And UV oxidation works on the principle of
21 applying ultraviolet light and also oxidants, ozone and
22 hydrogen peroxide. And the water enters into a baffled
23 reactor, and ozone bubbled up and through the bottom of the
24 reactor, and you have ultraviolet lamps that sit on the top
25 of the reactor. And those two, in combination -- and you

1 can also add hydrogen peroxide into the incoming stream.
2 And those oxidize your contaminants. And one very good --
3 one very good thing about this treatment is that you don't
4 produce a secondary waste like I was referring to with the
5 other two treatments.

6 For UV oxidation, we don't have any air emissions
7 other than -- well, we'll have just water. And we don't
8 have any emissions like we would get with some of our other
9 treatments that I'll be going over. One drawback to this
10 treatment is that it's the most expensive. And it also is
11 slightly operator intensive. You have to maintain the
12 chemicals at the facility.

13 For the third alternative, again, we have the
14 Hexavalent Chromium treatment, by ion exchange or by carbon
15 adsorption. And then we looked at an air-stripping
16 treatment. And, under this air stripping, we also looked
17 at two options within the air stripping. We looked at just
18 straight air stripping and also an air stripping and using
19 an activated carbon polish. And that activated carbon
20 polish is a similar treatment as we would use for the
21 Hexavalent Chromium.

22 And, for the straight air stripping, in order to
23 achieve our target levels, we would use two air-stripping
24 columns. And air-stripping towers work in the following
25 manner: The influent is dropped through the top of a

1 packed column. And you have plastic rings as a packing
2 material within that. And you force air up through the
3 bottom of the tower. And the application of air with the
4 water flowing over the rings causes volatilization of
5 organics. And that creates an emission through the top of
6 the tower.

7 And, for the State of Michigan, one thing that we do
8 have to consider is off-gas treatment. So we have to
9 implement a best available control technology. So what we
10 have done is selected a vapor phase carbon unit, which
11 would adsorb any of the volatile organics coming out of the
12 stripper. And then we can also regenerate on site using a
13 thermal oxidation unit. And so as soon as our carbon is
14 exhausted at the top of the column, they can regenerate it
15 through the thermal oxidizer. And the carbon is all ready
16 to go again.

17 Okay. The second variation of air stripping that we
18 have involves just one air-stripping tower again with the
19 vapor-phase carbon. And that's followed by a liquid carbon
20 adsorption treatment system. And that will act similarly
21 to the dual air strippers, will achieve the target levels.
22 And, with the carbon adsorption, we also produce a certain
23 amount of spent carbon through that process. And that's
24 approximately 200,000 pounds per year of spent carbon. And
25 that is shipped off site and replenished with a virgin

1 carbon.

2 And alternative four considered a steam-stripping
3 process. And I'd like to mention right now that the
4 remaining -- the air-stripping alternatives and the
5 steam-stripping alternatives have similar costs associated
6 with them. And the steam stripping is similar to an
7 air-stripping process. And the way it differs is that we
8 introduce the steam, which means we introduce heat which
9 causes the contaminants to volatilize more quickly than in
10 just a straight air-stripping process.

11 Now, with this we have no -- well, we do have a
12 secondary waste stream, but we don't have any air
13 emissions. As to what is done in the steam-stripping
14 process is: At the top of the column the emissions are
15 condensed back into a liquid form. So now we would have
16 concentrated organic waste that we would have to dispose
17 of.

18 I would like to go over a few of the operator
19 requirements. The stream-stripping process is somewhat
20 operator intensive and, as I mentioned, also is the
21 ultraviolet oxidation process. So therefore the
22 air-stripping alternatives are probably the least operator
23 intensive. However all of them do require a daily check
24 just to make sure that the system is maintaining itself and
25 take water samples every now and again for analysis.

1 Okay. To address the contamination at Kysor, we
2 developed six alternatives for soil treatment. And the
3 contaminated area behind Kysor is not at the surface level,
4 so it's not a direct-contact hazard. However, it does
5 contribute as a source to groundwater contamination;
6 therefore, that is why we're implementing a soil treatment
7 alternative. Now, the soil -- there's approximately 13,000
8 cubic yards of soil that would require removal. And the
9 contamination exists beginning at the 6-foot depth and
10 continues on down to about 25 feet.

11 And, again, we developed a no-action alternative
12 against which we can compare the other soil treatment
13 alternatives. And primarily the no-action for the soil
14 would be to establish a perimeter fence and also to
15 implement some deed restrictions limiting access to the
16 site. However, a fence does not limit access for animals
17 and is also subject to vandalism and other things.

18 And for the second alternative we developed the
19 containment alternative. That's the only containment
20 alternative. And that would be to institute a subsurface
21 containment wall and also a surface cap. And what the
22 containment wall does, -- this would be a circumference
23 containment wall -- it would go around the area of the
24 soils. It would leave the soils in place. And then the
25 cap would be placed over top to limit any infiltration. So

1 we would contain the area where the soils are, put the cap
2 on, therefore limiting the migration to the groundwater.
3 Now, for this alternative to work, we have to key or set
4 our containment wall down into an impermeable layer.

5 So at Kysor what we would do is -- Ted mentioned we
6 have a confining layer that's approximately 60 feet down
7 that we would use to key in. So we'd have the sides set
8 down into something stable. And typically a soil and
9 Bentonite slurry is used. Bentonite is best referred to as
10 a clay material. And the capping material would also be
11 like a two-foot clay cover. And above that we would have
12 two synthetic liners and a drainage layer and then a
13 vegetative cover. And, as I mentioned, this alternative
14 does keep the waste remaining on the site, so it is not a
15 permanent remedy.

16 For a third alternative, this involves excavating the
17 soils from in place and instituting a low-temperature
18 thermal-stripping unit. And thermal stripping is somewhat
19 similar to soil incineration that I'll be talking about as
20 alternative six. And it involves using a rotating drum in
21 which you place the soils. And the drum is heated up to
22 about 700 degrees Fahrenheit. And from that you obtain
23 volatilization of the contaminants within the drum, and
24 those can go into an after-burner, which is at a higher
25 temperature, to further destroy the organics. And some

1 particulates are drawn up through this drum and through the
2 after-burners. And those would be knocked out in a bag
3 house.

4 And for alternative four, it involves in-situ soil
5 flushing. And, for this alternative, this would be placing
6 horizontal perforated piping up in the unsaturated soil,
7 which means above the groundwater and also above where the
8 soil is contaminated. So that would be above the six-foot
9 depth, we'd place horizontal pipes with perforations and
10 then infiltrate water down through the pipes over the
11 contaminated area. Now, this would serve to draw the
12 contaminants from the contaminated area down into the
13 groundwater.

14 And there we would establish a series of groundwater
15 recovery wells whereby we would collect contaminants. And
16 then we could institute one of the groundwater alternatives
17 that I have gone over and put that extracted groundwater
18 from the soil-flushing system through the groundwater
19 treatment unit. However, for the soil-flushing
20 alternative, I do mention that we percolate water down
21 through the contaminated soils. So one of the drawbacks to
22 this alternative is: If there are preferential flow zones
23 for the water that are developed through the soil, we may
24 possibly miss collecting part of the contaminant through
25 the extraction well system. So that is one drawback to the

1 alternative.

2 Next is in-situ vacuum extraction. And this
3 alternative works on the principle of placing vertical
4 piping into the soil, down into the groundwater table which
5 exists at about the 25-foot depth. And this piping is a
6 slotted piping, and you force air down through the pipes
7 which are set in a grid pattern, and the grid pattern can
8 be established based on pilot testing as to the separation
9 between the pipes. So air is forced down, and then we have
10 extraction wells that are set in. They collect the
11 volatile organics. And forcing the air down into the soil
12 causes the contaminants to become desorbed from the soil
13 and collected up through a vacuum area.

14 Now, this also -- because this is drawing out volatile
15 organics, we do have to implement either a carbon
16 adsorption or an after-burner in order to maintain within
17 the emission standards for State of Michigan.

18 Our final alternative is incineration. And, as you
19 can see, it's the most costly of the alternatives, with
20 its similar treatment, thermal air stripping, being the
21 second most costly. And the reason for this alternative
22 being the most costly is probably because it is the most
23 effective alternative. And it also has somewhat more
24 complicated equipment. And the incineration unit acts
25 somewhat similar as an after-burner. We heat at very high

1 temperatures, preferably about 1400 degrees Fahrenheit.
2 We heat the soil in the burner area. And then through that
3 it goes through a bag house system to knock out the
4 particulates. And it generally can achieve a
5 99.9999-percent removal of all organics; thus, it is the
6 most effective of the alternatives.

7 However, we do feel that any one of these alternatives
8 could meet the groundwater -- excuse me -- the soil target
9 levels. However, as I mentioned, the soil flushing, there
10 is a question as to whether all of the contaminant would be
11 collected in the groundwater extraction wells.

12 Okay. At this time Dianne is going to go over the
13 preferred alternatives for both groundwater and for soil.
14

15 MS. ROYCRAFT: Thank you, Beth. These alternatives,
16 I'd like to mention, are described in the proposed plan
17 that's in the form of a fax sheet, copies of which were
18 sent to those on the mailing list. And we have some copies
19 also available here tonight. The preferred alternatives
20 that are preferred by both our department as well as the
21 EPA are, for the groundwater, a purge, treat and discharge
22 option.

23 And, for that option, we are recommending the
24 alternative 3A, which is the air-stripping system. And, as
25 Beth mentioned, that does consist of a dual system air

1 stripper which, with the influent blowers in, the
2 groundwater is directed down through the packed medium with
3 the objective of that packing material being separation so
4 the contaminants are actually separated from the
5 groundwater. And the blowers pump the concentrate into --
6 the air is directed upwards to an off-gas treatment system.

7 The second air stripper packing unit is used for the
8 final polishing of the contamination. This we feel is the
9 most cost-effective remedy available and has also the
10 advantage of not generating the volume of waste as some of
11 the other alternatives that would require off-site
12 disposal. And also it's less labor intensive.

13 For the soil cleanups, the alternative that is being
14 recommended is the in-situ vacuum extraction. And this is
15 a graphic depiction of what that system would consist of,
16 forcing air in through the ground and the volatilization
17 coming up through a series of pipes and then treated above
18 ground by a carbon adsorber unit. We feel that the
19 conditions in the Cadillac Industrial Park are particularly
20 favorable to this system. And it's a proven technology
21 that's readily available.

22 Those are our preferred alternatives. And at this
23 point I would like to again add that public comments are
24 available to be submitted through August 28th. And
25 comments that are submitted either tonight or through the

1 end of the public comment period will be responded to in a
2 responsiveness of a summary which will be mailed to all
3 those that are on the mailing list. And, for those of you
4 who are not on the mailing list and would like to be,
5 please indicate so to us.

6 The documents that have been generated during the
7 remedial investigation and feasibility study, some of which
8 are provided here tonight, are available in the repository,
9 which is the Cadillac Wexford County Library at 411 South
10 Lake Street here in Cadillac. We are available during
11 tonight's session and, of course, throughout the public
12 comment period to answer comments and questions. We'll be
13 available later tonight. And at this point I'd like to
14 introduce Steve Sanders from USEPA, and he will provide the
15 EPA perspective on this process. Steve.

16
17 MR. STEVE SANDERS: Good evening. I don't have any
18 pretty pictures. I'm just going to stand over here in
19 front of this microphone. EPA became involved in this
20 project when two of the industries in the Cadillac
21 Industrial Park were proposed for the National Priorities
22 List. Northernaire Plating was proposed in 1982, and Kysor
23 of Cadillac was proposed in 1985. When they're proposed
24 for the National Priorities List, this makes them more
25 commonly referred to as Superfund sites. And being

1 Superfund sites, that brought the EPA in through the
2 Superfund program involvement.

3 Our role in this project -- this has been a State lead
4 project all along. So our role has just been one of
5 oversight. The State has done the project throughout, and
6 we've just, you know, overseen and provided our comments
7 along the way. At this phase in the project, in
8 cooperation with the State, we've proposed a remedy to
9 clean up this contamination. And we're in agreement with
10 the State on the remedy that we've proposed. And, with
11 your help and your comments, we hope to provide a record of
12 decision, which is a decision provided by the EPA from the
13 regional administrator, on how these sites will actually be
14 remediated.

15 And I want to clarify. Our involvement is just
16 through the two Superfund sites. We're hoping to have this
17 record of decision completed sometime in mid-September,
18 once again, with your comments and your help.

19 So, in summary, really we're just -- EPA has just been
20 in an oversight role through this whole project. And we
21 agree with the State's -- well, in cooperation with the
22 State, we've provided our proposed alternative to clean up
23 these sites. And, with your comments, we'll make that a
24 final decision and proceed with the design and the actual
25 cleanup of the site.

1 That's about all I have. I'll turn it back over to
2 Dianne for your comments.

3 MS. ROYCRAFT: At this point I'd like to invite the
4 public to come forward and provide your comments. And,
5 again, we do request that you do sign in through the
6 sign-up sheet so that we do have the correct spelling of
7 your name. And public comments will be available through
8 the next three weeks.

9
10 MR. WALTER MEINERT: Thank you for the opportunity to
11 be here this evening and to comment on the RI/FS. My name
12 is Walter Meinert. I am a groundwater hydrologist and the
13 vice-president of the consulting firm of Fishbeck,
14 Thompson, Carr and Huber in Grand Rapids, Michigan. We
15 are quite familiar with the local and the regional
16 groundwater hydrology in the Cadillac area, having been
17 instrumental in the development of the Cadillac Municipal
18 well field in 1960. And we've also had the opportunity to
19 conduct several hydrogeological investigations for the
20 Kysor Industrial Corporation in Cadillac.

21 And we're here this evening representing the Kysor
22 Industrial Corporation. And we have been asked to review
23 the agency's RI/FS document and also to review the basis of
24 design and objectives of the Cogeneration Michigan, Inc.'s
25 facility. And, in doing that, we feel that the

1 Cogeneration Michigan, Inc.'s facility should have very
2 definitely been included as a remedial alternative on the
3 RI/FS. We think that the cleanup approach prepared and
4 performed by Cogeneration Michigan, Inc. is going to be
5 the same approach as the RI/FS intends with purge and treat
6 with air stripping with maybe just two minor differences.
7 You might consider them major differences.

8 But basically the purge system, as proposed by
9 Cogeneration Michigan, Inc., intends to clean up not only
10 specific contamination plumes but every contamination plume
11 in the Cadillac Industrial Park area. And I think we might
12 want to add at this point that, reviewing the document, the
13 FS states that there are six, I believe, contributors to
14 contamination which result in seven plumes. But at the
15 same time there's an inconsistency in the FS, whereas the
16 extraction systems are identified as Kysor Shallow A,
17 Kysor Shallow B and Kysor Intermediate.

18 And I think it's been documented well in the FS that
19 there are several other contributors to both the shallow
20 and intermediate plumes above and beyond Kysor. So, by
21 inference, that someone might take the position that all
22 the contamination in both the intermediate and shallow
23 plumes might be the result of Kysor's facility, where
24 indeed we know that's not the case.

25 I think that the only difference in the treatment -- I

1 mentioned that the purge system is going to contain all the
2 contamination, and it's going to contain all the
3 contamination by the proposed accelerated cleanup rate by
4 increased pumping rates. But the treatment system itself
5 has two basic difference from the proposed RI/FS. And that
6 is: The off-gas, whereas in your proposed plan is going
7 to be treated through carbon adsorption, the Cogeneration
8 facility will treat the off-gas through a thermal oxidation
9 to the boiler system.

10 And this has been documented by several of the largest
11 boiler manufacturers in the county, that this is very
12 technically feasible. We understand that the requirement
13 for the DNR is that you must have a detention time of one
14 second at 2200 degrees Fahrenheit. This can be
15 accomplished through thermal oxidation in a boiler. So we
16 think this is a very feasible and viable option that should
17 be considered.

18 And the other alternative -- not a separate
19 alternative -- but an efficiency of the Cogeneration
20 facility is that we propose to accelerate the cleanup
21 tremendously and accelerate the cleanup by increasing purge
22 rates. Cogeneration Michigan is proposing to pump up to as
23 much as 5,000 gallons a minute, whereas I understand in the
24 RI/FS the remedial program has been selected on the basis
25 of pumping 940 gallons per minute.

1 We have preliminarily at this time incorporated the
2 use of a two-dimensional model to show that increased
3 pumping rates at selected optimum strategic locations can
4 actually and effectively improve and increase the
5 accelerated time of capture. And what we intend to do at
6 this time is to incorporate the use of a coupled transient
7 flow solu-transport model so we can very, very
8 strategically and accurately locate purge wells, locate
9 pumping rates or determine pumping rates and determine
10 influent contaminant concentrations to our treatment system
11 such that we will be able to place wells at the right
12 locations, pump at selected rates and realize tremendously
13 increased accelerated clean-up periods.

14 So we think very definitely that the agency should
15 strongly consider the use of the Cogeneration facility to
16 clean up the Cadillac area groundwater contamination in a
17 very, very accelerated manner. I think that's all I have
18 this evening. And, with this, I'd like to introduce
19 Mr. Raymond Weigel, president of Cogeneration
20 Michigan, Inc. Ray?

21
22 MR. RAY WEIGEL: Thank you, Walter. I'd like to say
23 first that we of industry -- and I still consider that I'm
24 part of industry -- recognize the seriousness of the matter
25 of groundwater contamination and air contamination. I know

1 George Kempton, who is my successor over at Kysor, is a
2 fisherman, an outdoors man, and I don't think I have to
3 tell you how much interest Mr. and Mrs. Winn have in water.
4 That would go without saying.

5 I'm here to talk about Cogeneration. What is it?
6 Well, Cogeneration is producing two forms of energy with
7 one type of fuel. We are going to produce here 45
8 megawatts of electric power. And, if I can put that in
9 perspective for you, what does Cadillac use?

10 Seventeen-and-a-half megawatts, roughly. So we're going
11 to produce more than twice what the entire City of Cadillac
12 will use. We're going to also produce 35 pounds of steam
13 per hour for industrial use. We're going to burn wood in
14 this facility, refuse wood -- underline refuse wood -- per
15 year, 500,000 tons.

16 We're going to have a lumber mill in connection with
17 the facility producing 50 million board feet of
18 commercial-grade lumber annually. That will require
19 125,000 cords of wood, fundamentally red pine. In the
20 forrest we will get, from cutting that red pine, about
21 200,000 tons of waste. The sawdust, the bark and the scrap
22 from the mill will provide about 150 tons. The other 150
23 tons we will gather from the forests in the area. An
24 interesting statistic for you might be that my studies have
25 shown -- and they're confirmed by the DNR -- that in

1 Wexford County alone there is a renewed -- annual renewable
2 amount of waste wood of 1,131,400 tons. Now, remember I've
3 said we're going to burn 500,000 tons. So there is more
4 than two-and-a-half times that much available in Wexford
5 County alone in waste wood.

6 Most of the timber and the waste, before I'm attacked
7 by some so-called conservationists, is going to come from
8 land that we will own, 6700 acres, or land that we will
9 have under contract to timber, 22,000 acres. The total
10 cost of this project is 102 million dollars. I'm delighted
11 that the approval was received from the banks to go forward
12 on the 17th of July. And, as many of you know, I held a
13 press conference on the 21st of last month.

14 We've had a contract with Consumers Power since
15 February of 1987, and we have been delayed because of the
16 problems of the Michigan Public Service Commission and the
17 consumers' battles with the attorney general and the group
18 called ABATE. We have all of the clearances that we
19 require at this point.

20 Groundwater cleanup has been a part of this project
21 since I first became involved. And I looked in my file the
22 other day, and it was interesting to me to note that on
23 March 3rd, 1984, our mayor, Daryl Becker, sent me a little
24 note and said, "I thank you for agreeing to serve on the
25 Cogeneration Advisory Committee," which I did. And I made

1 a few suggestions. And you know what happens when you
2 make suggestions? You become chairman. And unfortunately
3 or fortunately that's what happened to me. And then I
4 became president. And, incidentally, I was paid last year
5 my dollar for 1987. I haven't gotten the rest.

6 You will recall also that on the 12th of September, I
7 believe it was, in 1988, I appeared before the EPA in
8 Chicago, and I presented the story of Cogen and what we
9 hoped to do. And now we're reality. And this is what
10 we're going to do. We'll be spending, in addition to what
11 has already been done -- and the good work that you folks
12 have done. And I can't say too much for that. We're
13 spending seven-and-a-half million dollars for capital
14 equipment to do groundwater cleanup.

15 Now, before anyone asks the question, "How much is
16 that going to cost industry for that capital equipment,"
17 let me answer: Nothing. Louie Fernandez, former chairman
18 of the Monsanto Chemical Company, came over to see me one
19 day. Louie has a home on Crystal Lake now. His
20 headquarters used to be in St. Louis. He was the man that
21 started Clean Sites, Incorporated, in Washington, D.C.
22 Clean Sites was financed by Allied Chemical, Dupont,
23 Monsanto, Dow, because they -- and it's purpose is to find
24 a way to solve these groundwater problems. You're probably
25 all familiar with it.

1 And the Superfund was one of the first efforts in that
2 direction. But I remember so vividly four years ago
3 reading an article where Lou Fernandez talked about clean
4 sites, and he talked about the Superfund, and he talked
5 about cleaning up contaminated groundwater. And, at that
6 point, the EPA had indicated there were 1200 critical dump
7 sites in the United States. 1200! Now I think you number
8 them -- what? -- 30,-, 40,-, 50,-, 100,000. But he said at
9 that point four years ago, "The industry and government
10 together have been able to clean up 36 critical dump sites
11 under Superfund." And he called that glacial progress.
12 And I would agree. That's glacial progress.

13 It's not government's fault. It's not industry's
14 fault. It's that we haven't gotten together to do the job,
15 and that's what we've got to do. I think here the City is
16 the issue. Not Kysor, not Four Winns, not Rexair, not
17 Brooks Perkins. I think the City is the issue. You have
18 an industrial park, the Harry VanderJagt Industrial Park.
19 That's where this is going to go. But, if you can't sell a
20 vacant lot in the industrial park, forget it. What good is
21 it? And, if it's contaminated, I can tell you from the
22 experience of industry, and I think Mr. Winn sitting here
23 and Mr. Kempton will tell you that they wouldn't buy a
24 piece of property unless they checked it out very carefully
25 from this point on.

1 So the important thing is the City. Mr. Meinert
2 talked about our pumping up to 5,000 gallons of water a
3 minute. And that's what we intend to do. We're going to
4 run pipes out -- we're just beyond Leeson. If you wonder
5 where that is, the Ann Arbor railroad will be out in back,
6 that line along there, in the back of the plant
7 (indicating). And Leeson is just up ahead of it. 13th
8 Street is about where the "C" in Cogeneration is. We're
9 going to bring the water from the wells.

10 And we agree with your number, by the way. It's very
11 interesting. We think you're awfully bright, because
12 Walter Meinert came up with nine, but he didn't include the
13 Hexavalent Chrome, Tom, and nor did he include the Haring
14 Township. So we add those two that you added. And we're
15 in agreement with eleven.

16 I would hasten to add that this group didn't suggest
17 5,000 gallon a minute. And I'll tell you why they didn't:
18 Because they knew that that would cost industry one big,
19 bloody bundle, and they were doing a conservative job. And
20 I commend you on that. But we're sticking our chin out.
21 And we're saying that we'll go up that high. We can do it,
22 because we have the basis for doing it. A cogeneration
23 facility requires a lot of water. And so we're going to
24 use that 80-million-dollar plant that does a few other
25 things on the side, and we're going to use it as an

1 incinerator for contaminated air.

2 We're going to have a air-stripping device, as you
3 suggested. We agree thoroughly. We're going to take those
4 17 known contaminants out of the water. We're going to
5 burn them at 2200 degrees Fahrenheit. And the residuals
6 from all those, because they are, in the main, oil
7 products, will be H2O, HCL and Co2. Co2, carbon dioxide.
8 The trees, the grass love it, and we're going to emit that
9 out of the stack. H2O will be steam. Now, check your
10 chemistry of high school. The HCL we're going to
11 neutralize with NAOH, sodium hydroxide. I'll get there in
12 a minute. And we'll have H2OHNACL (sic). Right?
13 H2ONACL. Right. Salt water. And that we'll put in the
14 river someplace or the Clam or wherever you decide -- the
15 DNR decides that it should go.

16 We believe that we're going to do this a lot more
17 quickly -- we're sure we are -- than has been suggested
18 here, because we're going to use much more volume -- much
19 higher volume of water and because we have the capability
20 of using some different sciences when we get up to that
21 volume. As some of you know, I spent a few years at Kysor,
22 42 so far, and I've spent 25 as a director of Mercy
23 Hospital, 15 as chairman. And I started -- helped work on
24 a bank. Bill, do you remember that? And I was a director
25 of Spartan. I guess I brought WWTW to Cadillac, and I

1 built the first medical arts center here. And I got
2 involved in a few other things around town.

3 But this has been the most frustrating, demanding,
4 tiring, interesting, educational and now rewarding thing
5 that I've ever done to now. But, when we get this going,
6 I've got some other projects. Thank you very much.

7 MS. ROYCRAFT: Thank you, Mr. Meinert and Mr. Weigel.
8 I would like to comment that in reference to why
9 Cogeneration was not named in the feasibility study, is
10 because of the explanation given. It is more than a
11 cleanup system. And the focus of the feasibility study is
12 on the cleanup of the groundwater and soils based on
13 investigation; the lumbering operation, the generation of
14 steam and electricity will not ultimately be funded at tax
15 payers' expense. And that is why Cogeneration, as such,
16 was not identified in the feasibility study.

17 We certainly agree that that project considers
18 strong -- that merits consideration, and we have had an
19 open dialogue with Cogeneration's consultants, and we have
20 shared all of the documents generated during this process.
21 And we do look forward to the proposal for Cogeneration. I
22 would like to ask: Is ground breaking still scheduled for
23 Friday?

24 MR. WEIGEL: No. We're having to put that off Dianne.
25 We have had a few complications, some interesting ones,

1 arise. There's some problems. I spent all day Thursday
2 and Friday in Ann Arbor with our contractors. And it seems
3 that we're a little apart, like 7 million dollars, on the
4 budget. And so we're doing a little -- we're going to have
5 a few more prayer meetings before we schedule our ground
6 breaking. Additionally, we have, as of the moment, about
7 eight major firms who would like to take an equity
8 position in Cogen, including Consumers Power Company.

9 And I am currently negotiating. And I mentioned to
10 you that we're going to have lands of our own. Consumers
11 have now agreed to sell us 6732 acres on the Manistee
12 River. They have also agreed to work out a program for us
13 to timber 22,000 acres that they own in Michigan. So we're
14 going to work on that basis with your forestry department.
15 And I don't want to get things moving too quickly until we
16 get those put to bed really.

17
18 MR. DON RENNIE: I have two letters that I'd like to
19 read. For the record, my name is Don Rennie. I'm with the
20 City of Cadillac. And the first letter is from Daryl
21 Becker, mayor of Cadillac:

22 "I am unable to attend the public hearing to
23 discuss the recommended cleanup action for the
24 industrial area located in the northwestern part of
25 Cadillac. Cadillac City Council will be meeting in

1 its regularly scheduled council meeting at the same
2 time as your hearing. I would like to address the
3 problem of groundwater cleanup, and I have instructed
4 the City's utilities director, Donald Rennie, to
5 represent the City in this regard. In addition, I
6 would request that this letter be entered as part of
7 the record.

8 Over six years ago, the City of Cadillac
9 recognized that there existed a problem of groundwater
10 contamination in the industrial area of the City. At
11 that time, we began a feasibility study to determine
12 if a process utilizing a wood-fired cogeneration plant
13 in conjunction with an air-stripping device could
14 clean up the water in the entire area and do so at an
15 affordable rate. Kysor Industrial Corporation soon
16 became a partner in this project. And we have finally
17 moved to a point where we are ready to break ground on
18 the project.

19 We would have been operational if the Public
20 Service Commission of Michigan had not held up the
21 final approval during a protracted hearing on
22 cogeneration in Michigan. Now that that is past, we
23 are prepared to move forward. The City is preparing a
24 method to insure that there is adequate funding for
25 the groundwater cleanup portion of the project. The

1 City has invested funds as seed money to bring this
2 project to this point in time, is a full partner in
3 the project and will be providing the land in the
4 City's industrial park.

5 This problem is a Cadillac problem. We believe
6 we have the means to solve the problem; therefore, we
7 request that the EPA and the MDNR work with us by
8 recognizing the Cogeneration project is the most
9 expedient and preferred means of handling this
10 environmental problem."

11 The next is a statement by the City, and it goes as
12 follows:

13 "Nearly 11 years ago the City of Cadillac became
14 aware that groundwater was being contaminated by a
15 plating operation near the City well field.

16 Subsequently several other contamination plumes have
17 been identified in the same general area of the City,
18 all of which have the potential to contaminate the
19 public water supply. During the ensuing 11 years, a
20 large amount of effort and money has went (sic) into
21 evaluating this admittedly complex problem.

22 Unfortunately very little in the way of real cleanup
23 has taken place. And the City water supply, while
24 showing no sign of contamination at this time, remains
25 in the same relative position it was in 11 years ago.

1 The City of Cadillac has three very simple goals
2 that are related to this process: First, to insure
3 that our public water supply is protected from
4 contamination; second, to see that the effective
5 cleanup of existing contaminants take place as rapidly
6 as possible and to do so while retaining existing
7 industry and jobs they represent, along with
8 encouraging continued industrial, commercial and
9 community growth.

10 The City believes that the cogeneration facility,
11 as proposed by Cogeneration Michigan, Inc., will be
12 able to accomplish the task of groundwater cleanup for
13 the study area in such a way that it will meet all of
14 the City's goals and do so in a complete and
15 cost-effective manner. It most certainly will be able
16 to complete the cleanp process in much less than the
17 64 years proposed in the feasibility study under
18 consideration at this time.

19 In addition, using cogeneration as the method of
20 choice for cleanup will not cost the taxpayer anything
21 as compared to millions of dollars in capital and
22 operating costs contained in the feasibility study.
23 To begin the process of cost allocation, Cogeneration
24 Michigan, Inc. has hired a consultant to prepare a
25 special assessment process to spread the collection

1 and treatment costs between the several industries
2 which will be involved in the cleanup. The City of
3 Cadillac remains committed to a speedy and yet
4 cost-effective process of groundwater cleanup. We
5 believe that the planned cogeneration facility has the
6 best opportunity of reaching those goals."

7 Thank you.

8
9 MR. DAVID CROOKS: Good evening. My name is David
10 Crooks. I'm appearing on behalf of Kysor Industrial
11 Corporation. As Mr. Weigel indicated, he's our past
12 chairman. Now Mr. Kempton is in that position. So I'm
13 here to speak on behalf of Kysor Industrial. Kysor has
14 extensively, over the years, studied the E.C. Jordan plan,
15 the RI, the FS and now it's plan. My main message tonight
16 in way of my comments will be very brief. Kysor will be
17 submitting written comments within the written comment
18 period.

19 Kysor shares the interests of the DNR and the EPA in
20 addressing the environmental issues associated with the
21 Cadillac Industrial Park and the Cadillac groundwater area
22 contamination site. Predating the RI/FS and the
23 undertaking by the DNR and E.C. Jordan, Kysor, in 1981,
24 excavated over 700 cubic yards of soil behind its plant.
25 We, at the request of the DNR, did a two-phase

1 hydrogeological study, quite extensive I believe, and part
2 of which -- or a large part, I think, incorporated in your
3 plan conducted by Mr. Walter Meinert, who you've heard
4 from earlier.

5 That plan or study commenced in 1982. The first
6 phase was reported in 1983, the second phase in 1985.
7 Those studies concluded that there was no threat to the
8 City of Cadillac well field. And, at that time, Kysor
9 recommended that the numerous monitoring wells drilled as
10 part of that study be monitored. At just about that same
11 time Cogeneration Michigan came on to the scene. My main
12 message here tonight is to reflect Kysor's strong support,
13 and echo that of the letters from the mayor and the City
14 Council for and on behalf of Cogeneration.

15 As you know, -- and, Dianne, you commented -- Kysor,
16 over a year ago, wrote a letter to the two agencies and
17 urged that cogeneration and, in particular, its
18 groundwater features be addressed as part of the
19 feasibility study. While we were disappointed and do
20 understand, if not accept, the explanation, it was not
21 considered as part of that study. Nonetheless, today I
22 want to reassert that Kysor is willing to work with the EPA
23 and DNR towards an effective alternative. We believe that
24 alternative is cogeneration. We would, based on that
25 statement, however, request that the EPA and DNR work with

1 us in order to implement and to work together, pushing
2 towards that same goal that cogeneration be the chosen
3 option to clean up groundwater in our City. Thank you very
4 much.

5 MS. ROYCRAFT: Thank you, Mr. Cooks. Again, I would
6 like to emphasize that we don't disagree that Cogeneration
7 merits consideration. And, because it has not been named,
8 per se, in the feasibility study, that does not mean it is
9 precluded from consideration. And we will entertain the
10 proposal when it is submitted.

11 MR. COOKS: Thank you.

12 MS. ROYCRAFT: Any additional comments? Please come
13 forward.

14
15 [REDACTED]: My name is [REDACTED]. I'm
16 just an ordinary citizen. Now, I feel rather snowed under
17 here. I have been interested in this matter of water
18 contamination such as aquifer at least a dozen years. I
19 wondered about the effects of some of these volatile
20 organic compounds. I spent about 18 years at an automotive
21 engineering laboratory where I was quite a bit exposed to
22 several of these compounds that are mentioned, probably
23 some that are not mentioned.

24 And I have yet to find a doctor who says that they're
25 really so very harmful. So I just wonder how important

1 some of this is. Really, what is the risk? I don't get a
2 clear picture of what the risk is with these compounds.
3 Some people say they're very deadly, and others say that
4 they don't amount to anything. There are acute effects of
5 some of these compounds, meaning that they hit you right
6 now. And then there are also latent effects which are
7 continuous over a period of time, small exposure for some
8 period of time.

9 But, like I say, I'm puzzled by some of the answers
10 that I get when I ask questions. In some cases,
11 effectively I've been more or less told to mind my own
12 business. But I just don't know how to feel about some of
13 this. But I do know that some of these compounds are about
14 one-and-a-half times the weight of water. So, if you get
15 some trichlorethylene, perchloroethylene, something like
16 that, in the ground or if you get it in water, a body of
17 water, I don't see how it can be evaporated from the
18 surface of the water, because it's going to be under the
19 water.

20 The thing is, when -- I learned some time ago that the
21 City of Charlevoix switched from wells for their water
22 supply to Lake Charlevoix. Now, I never did get a clear
23 picture of what the setup is there, but I assume that these
24 compounds are settling to the bottom, and they're picking
25 the water off at some intermediate stage so they're not

1 picking up these volatile compounds. But I never, like I
2 say, have had a real clear picture of that.

3 There was a case where there was a spill of dry-
4 cleaning fluid near Baldwin. It contaminated several wells
5 including the water supply of a restaurant. The Lake
6 County Board of Commissioners, apparently from their
7 minutes, discussed this situation. But it was couched in
8 such language that it wasn't clear. A person had to read
9 between the lines. And a county commissioner told me that
10 that was a local affair and shouldn't interest the general
11 public.

12 Now, I don't feel that's a right attitude. And my
13 employer in the Flint area told -- I was told by somebody
14 in the management that the waste that we were producing,
15 the used solvents, were being recycled. And later it
16 appears, from what came up, a lot of that was put into a
17 landfill and the Swarts Creek area.

18 So I don't know how to feel about some of these
19 things, but I certainly would like to see some prevention
20 initiated along the line, moreso than what there is now
21 with some of these compounds, and let the public know a
22 little bit more about what is going on. When I am told
23 that this is a local affair or something like that, I live
24 between Cadillac and Baldwin, so I visit both places,
25 different ones. If I go to a restaurant and drink polluted

1 water, I think that should be of interest to me and the
2 general public. But I guess that's about all I can say at
3 the moment. Thank you.

4 MS. ROYCRAFT: Thank you. [REDACTED], is it? Do we
5 have the correct spelling of your name?

6 MR. SANDERS: He's writing it down now.

7 MS. ROYCRAFT: Do we have other comments tonight? Any
8 questions that should be entertained at this time?

9 MR. [REDACTED]: Yeah. I'd like to ask a question.

10 MS. ROYCRAFT: Okay.

11
12 [REDACTED] [REDACTED]: My name is [REDACTED]. I think the
13 cogeneration project is a great project. But I would like
14 to know what connection cogeneration has to clean up the
15 contaminated area in the industrial park. I actually don't
16 see no connection.

17 MR. WEIGEL: Interesting question.

18 MS. ROYCRAFT: [REDACTED] [REDACTED], would you put your name on
19 the list, please, when [REDACTED] gets done? Right in
20 front of you.

21 MR. WEIGEL: Let me go back in history for you a
22 little bit. Some years ago Walter Meinert, who spoke to
23 you earlier in the evening, was in Cadillac working with
24 Kysor on the matter of contaminated groundwater. And I was
25 invited to attend a luncheon with he and Mr. Kempton and

1 several others. And I had retired at that point, I
2 thought. And Walter mentioned, during the course of our
3 luncheon, that he was working with the City -- you know, he
4 and his firm designed the City wells, by the way -- and
5 happened to mention also that he was talking to the City
6 about a cogeneration facility and did I know anything about
7 it? And I said, "No, I don't." And I should have stopped
8 right there, but I didn't.

9 But he said, "Well, I just wondered, because it takes
10 tremendous volumes of water to handle a cogeneration
11 facility." And I said, "What do you mean?" And he said,
12 "Oh, it could be 1,000, 1500, 2,000 gallons a minute."
13 And, to me, that's a lot of water. And that evening I got
14 thinking about it. And the next day I called Walter, and I
15 said, "Walter, you talked about a lot of water. Can we
16 pump that much water in the Cadillac area?" And he said,
17 "Yes, you can, because you're in the Muskegon River Basin."
18 I believe that's what you told me, Walter. "And you have a
19 tremendous replenishment capability in the industrial park.
20 And, yes, you can pump maybe up to as much as 7,000 gallons
21 of water a minute."

22 And Bob Hamilton, our City Manager, has told me he's
23 pumped all seven wells at the City park at one time. So we
24 can pump a lot of water out there. So I said, "Okay. If
25 that's the case, why don't we pump out the contaminated

1 water and treat it and do it in connection with the
2 cogeneration project?" The point being that you wouldn't
3 build a hundred-million-dollar contaminated water clean-up
4 facility. But, if you're going to build a cogeneration
5 plant anyway, why not put the two together? That was my
6 idea. That's that thing I told you, when you make a
7 suggestion and become chairman.

8 MR. KEMPTON: You might point out on the picture where
9 the air-stripping --

10 MR. WEIGEL: Well, actually, George, it's on the other
11 side of the building. There are four towers. You're
12 familiar with the blue silo that you see in farmyards?
13 It's made by A.O. Smith in Milwaukee, and they can be up to
14 18 foot in diameter and 70 foot tall. And what we will be
15 doing is bringing the -- with your approval and
16 concurrence, bringing the contaminated groundwater into
17 these four towers. It will be just on the other side of
18 this boiler house. That, by the way, is 90 foot tall. And
19 we'll be introducing the water at the top. It goes through
20 a series of baffles, plastic baffle arrangements.

21 And at the same time, just exactly as the DNR picture
22 shows you, if you take a look at the picture where it
23 shows a sirocco-type impeller blowing up through the bottom
24 of a big tank, that's exactly what this is going to be
25 doing. It will blow 25,000 cubic feet of air a minute

1 through each one of these containers, tanks. And that air
2 will come off of the top. And we have proven conclusively
3 that we can take 99.97 percent of the contaminants out of
4 the water. And we've done this. We've shared all this
5 information with the DNR. I believe you'll concur with
6 that. Mr. Bails worked with us first some six years ago.

7 We will then introduce that contaminated air into the
8 boilers, again, at 2200 degrees Fahrenheit, and we burn it.
9 And then I talked about the residuals.

10 [REDACTED]: Can I ask one question?

11 MS. ROYCRAFT: Please.

12 MR. WINN: You're picking up the contaminated water
13 at a different level than the drinking water. No one said
14 that.

15 MS. ROYCRAFT: At a different depth.

16 [REDACTED]: That's right. He's picking up the
17 contaminated water that's never been deep enough. That's
18 the reason our wells hadn't been affected. So he's not
19 going to be picking up our drinking water and putting it
20 through there. He's going to be picking up the
21 contaminated water at the shallower level.

22 MS. ROYCRAFT: Right.

23 [REDACTED]: No one said that today.

24 MR. WEIGEL: About 140 feet, [REDACTED].

25 [REDACTED]: Where we're getting our drinking water is

1 deeper and is protected by 34 feet of clay.

2 MR. WEIGEL: 307 feet is the average depth.

3 MR. TAYLOR: I mentioned that when I first --

4 [REDACTED]: Oh, did you?

5 MR. TAYLOR: Yeah. That there was three levels of
6 that.

7 [REDACTED]: Well, I'm just making it clear to him.

8
9 MS. MARY STARK: I'm Mary Stark. And I probably will
10 be sending a letter off within the response time. I'm the
11 Haring Township Supervisor, and now I've got questions that
12 I didn't have earlier, after Mr. Weigel spoke. Just for my
13 own clarity, I guess, I now have a question with respect
14 to: It appears that someone has to approve his proposal
15 for the cleanup aspect of it. What happens if that
16 approval is not given? Will the amount of water that you
17 were proposing, 5,000 gallons a minute, will that be
18 reduced for the operation of the Cogen plant?

19 MR. WEIGEL: Drastically.

20 MS. STARK: Okay. And then my other question is: And
21 [REDACTED] brought this up. There are a number of people
22 that are pulling their drinking water from private wells
23 that are far above that 140-foot depth. And I guess I'm
24 wondering, through all the studies and everything that have
25 been conducted, is there a radius from this spot that

1 you'll be pulling from that you know you'll be pulling from
2 and possibly be drying some people's groundwater wells up?
3 Because I know that in the area there are wells that people
4 are drawing from that are 85 feet deep. And I'm just
5 curious what the studies are?

6 MR. WEIGEL: I'll defer to my learned colleague here.

7 MR. MEINERT: That will definitely be taken into
8 consideration in the design of a purge well containment
9 system. We have to purge enough water to totally encompass
10 the plumes of contamination. And that can be done at
11 lesser pumping rates than what myself and Mr. Weigel have
12 mentioned. However, when we select a pumping rate, it will
13 be such that it can accomplish the objective of cleaning up
14 the contamination at accelerated rates. At the same time,
15 it's not the intention to de-water the nearby residential
16 wells.

17 MR. WEIGEL: What Walter is talking about here -- if I
18 may, Walter?

19 MR. MEINERT: Go ahead, Ray.

20 MR. WEIGEL: We may pump 600 gallon a minute here, 150
21 gallon a minute here. And we may say on Tuesday we're not
22 going to pump that well, but on Wednesday we're going to
23 pump the tar out of that one (indicating).

24 [REDACTED]: Well, the wells that she spoke of would be
25 contaminated.

1 MR. WEIGEL: Beg your pardon?

2 [REDACTED]: The wells she's speaking of, if they're
3 getting the -- you know, dried up their wells, it's
4 contaminated water they're drinking.

5 MS. DeSWARTE: Not necessarily.

6 MR. WEIGEL: No, no, no.

7 MR. TAYLOR: It's outside the --

8 [REDACTED]: Well, they're going to only suck it out of
9 contaminated plumes.

10 MR. WEIGEL: Remember, these are roughly 307 feet.
11 Right, Tom? Around 307. And there are seven of them out
12 there. I think six are active at the present time. And
13 it's interesting that -- what did you find in your studies?
14 About 30 feet? 28 to 30 feet of clay here?

15 MR. TAYLOR: In that area. But it thins very quickly.

16 MR. WEIGEL: It pinches off this way?

17 MR. TAYLOR: Very quickly. By the time you reach the
18 Rexair building, it's essentially gone.

19 MR. WEIGEL: But all of the test wells that they've
20 done give us the information that Mr. Meinert was talking
21 about. And this is the reason for the three-dimensional
22 study that he was also talking about, that we'll do. And
23 it will be constantly monitored. We'll have CRT's out
24 there, truly, and looking at them. And that one is pumping
25 so much a minute, and that one is pumping so much, and

1 that's pumping so much, and that's pumping so much. This
2 is bringing in X-contaminant. This one is bringing in
3 nothing. We're going to start varying these things.

4 I wish I had a picture of the control room that's
5 proposed for Cogeneration Michigan. It goes from here to
6 the wall. A beautiful thing. Four colors -- CRT's. It's
7 very interesting.

8 MR. ROYCRAFT: Thank you.

9 MR. MEINERT: Ray, let me just expand on that a second
10 here. If we were to just increase pumping rates at random,
11 we very possibly could be affecting residential nearby
12 domestic wells. But, through the model that I've alluded
13 to initially and Ray spoke about, this is a coupled, solid
14 solu-transport transient state model. We're going to show
15 exactly what the impact and effect will be by properly
16 placing wells with selected optimum pumping rates.

17 If we were to take the wells as shown right as they
18 are in the RI/FS and simply increase the pumping rate of
19 those wells, we are going to actually extract clean water
20 at a ratio in excess of the dirty water. So the
21 contaminated water will not result in accelerated cleanup,
22 because you take any particular well, a higher portion of
23 that discharge will be clean water. But we can show
24 through some analytical approaches and numerical models
25 that if you optimally place where these wells will be, we

1 can restrict the zone of interference, but at the same time
2 accelerate the velocity and movement of contaminated water
3 to the well. So that's why this model that we have both
4 spoke of is extremely important and very effective to show
5 what the net effect will be.

6 MR. WEIGEL: Between Rennie and Meinert, I'm getting
7 to learn a lot about water.

8 [REDACTED]: Once you get the go ahead on this, how many
9 years will it take before you'll be operational?

10 MR. WEIGEL: We have the go ahead. We're just doing
11 our final negotiation with contractors. Our bankers have
12 given us the go ahead. And, you know, when your bankers
13 tell you you're all set, I think you'd agree that you're in
14 pretty good shape. So we have commitment now from the
15 banks Goldman Sacks, Smith Barney, Robert W. Baird and
16 Company, Northwestern Mutual.

17 [REDACTED]: Okay. How many years before you'll be
18 operational?

19 MR. WEIGEL: Twenty-four months.

20 [REDACTED]: Two years. Thank you.

21 MR. WEIGEL: Which roughly is the time the DNR feel it
22 would take, their optimum approach. A little less. But,
23 on the other hand, we're saying that we will have this
24 finished long before your grandchildren are worried about
25 drinking it.

1 [REDACTED]: My name is [REDACTED]. And,
2 you know, I think that we all realize that we have a very
3 severe problem in our industrial park. And certainly the
4 health hazards may be part of them, a big part of them. An
5 offshoot of this park -- of this problem, though, is the
6 salability of property, of people willing to come to town
7 and invest in property. We have properties like the Four
8 Star building which, as far as salability or anybody going
9 in there, has AIDS. We have other properties that -- maybe
10 the building dated the Four Star building because people
11 suspect it has AIDS, and no one will touch it.

12 So we end up with a situation where we have people
13 that are willing to come to town to make an investment but
14 are unwilling to take the chance of betting their whole
15 company or betting their capital on the liability of some
16 property. So we know that it has to be cleaned up. Now,
17 I'm just a farm boy from outside of town. And some of the
18 numbers that you put up there, you know, sort of boggle my
19 mind, especially when we talk about the no-action plan
20 costing almost a million dollars.

21 I'm concerned about staying around to see if your plan
22 really works. And, you know, we're looking at 64 years, I
23 believe, it was in there. And I really encourage you to
24 look carefully at Cogen and see if it won't fit into your
25 plan and see if it won't accelerate the cleanup period so

1 that I can be around to make sure that you guys did your
2 job right. Thank you.

3 [REDACTED]: There are a couple points that I wonder
4 about. What is the migration of these VOC's in dry soil?
5 In other words, since they're heavier than water, do they
6 tend to still continue to go downward, or do they tend to
7 stay stabilized in dry soil? What I was referring to
8 awhile back was the fact that some of these compounds are
9 heavier than water. And, if you have them in the ground
10 and you get water on top, the water is going to drive the
11 materials further into the ground. In other words, if you
12 cap the area like one of the ideas was, that would prevent
13 that. But that's one of the things that I wonder, is how
14 much migration is there in dry soil?

15 Another thing is, some time ago there was a spill of
16 solvent at a Cadillac industry, factory or whatever. And
17 there was an article in one issue of the Cadillac Evening
18 News. And after that there wasn't a word again in the
19 paper that I ever saw. Some time later at a meeting
20 similar to this I talked to a DNR man and got some facts on
21 what had happened, why that spill had occurred and such as
22 that. But, like I mentioned before, I'm concerned about
23 the way the public is misinformed or uninformed. Thank
24 you.

25 MS. ROYCRAFT: In response to your question on

1 migration of volatile organic compounds in dry soil, it is
2 true that lighter fractions may volatilize and move up
3 through the soil and the soil pore space, or some materials
4 may adhere to the soil particles particularly at the soil
5 water interface and be flushed down with the water with any
6 precipitation. So the interface at the soil and
7 groundwater may actually be a condensed zone of
8 contaminants. Do you want to comment any further on that?
9

10 MR. MARK MARENTETTE: My name is Mark Marentette from
11 the Evening News here in Cadillac. First of all, do I
12 understand that the estimated cost of your preferred
13 option is about 17 million? Is that in the range?

14 MS. ROYCRAFT: Yes.

15 MR. MARENTETTE: Has it been decided how that is going
16 to be paid for? Is that going to be charged to the
17 businesses who are found to be part of the contamination,
18 or is that going to be paid through tax payers' dollars?

19 MR. SANDERS: That would be something that would be
20 negotiated. We have a negotiating period, and that's when
21 they would entertain a proposal, this Cogeneration, I
22 assume. And it really depends on how the negotiations go.
23 But there is money available, you know, in the Superfund to
24 fund this type of project.

25 MR. MARENTETTE: Okay. Is there -- and I'm not sure

1 who would answer this question -- a significant
2 environmental difference between your preferred plan and
3 the Cogeneration method of cleaning up the groundwater, the
4 impact on the environment when all is said and done?

5 MS. ROYCRAFT: I think that's difficult to answer
6 without that proposal being submitted at this time.

7 MR. MARENTETTE: Thanks.

8 MS. ROYCRAFT: I should mention, also, after the
9 public comment period there, again, is a responsiveness
10 summary which does address the comments submitted
11 throughout that comment period. And the parties that have
12 been identified as responsible for the contamination will
13 be invited to negotiate for a cleanup alternative. And
14 the record of decision, which, as Steve mentioned, is the
15 decision-making document, is scheduled for yet this fiscal
16 quarter, being mid-September. So that's the time frame
17 that we're looking at for having the final decision on the
18 cleanup option to be implemented. Any additional comments
19 tonight?

20 MR. CROOKS: I have just a quick question? If one
21 desires a transcript of this proceeding, how does one get
22 it?

23 MS. ROYCRAFT: That will be available through EPA's
24 Office of Community Relations. Dan O'Riordan is the
25 Community Relations Coordinator at EPA. And his address

1 and phone number are provided on the proposed plan fax
2 sheet.

3 MR. CROOKS: Thank you.

4 MS. ROYCRAFT: If there are no other comments or
5 questions, we'd like to adjourn at this time. And, again,
6 we welcome and encourage your comments and invite you to
7 stay after to address us informally on this project.
8 Again, thank you.

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RECORDER'S CERTIFICATE AND NOTARIZATION

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Ann M. Holmes

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Ann M. Holmes, CER-2629
Notary Public
My commission expires 4-13-91